

Project Design Phase

Proposed Solution Template

Metric	Details
Date	08 July 2026
Team ID	K. Bhargav
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Clinical care teams and diagnostic specialists deal with massive, scattered multi-variable patient diagnostic datasets. Without clear, visual tools to crossreference overlapping factors (like age, smoking habits, diabetes, and stroke history), identifying high-risk cardiovascular clusters manually is slow, tedious, and prone to cognitive visual overload.
2.	Idea / Solution description	The proposed solution utilizes a clinical health dataset of 4,494 patient records across 18 features to build an interactive Tableau Dashboard and a 7slide Tableau Story. These visualizations cover demographic risks, behavioral impacts, and deep metabolic profiles. They are embedded into a custom lightweight Python Flask-based web application, giving cardiologists a centralized, accessible interface to explore medical insights through interactive filters.
3.	Novelty / Uniqueness	Rather than presenting static medical spreadsheets or relying on cluttered boilerplate corporate templates, the platform offers a focused, custom-built analytical web interface. By combining a clean HTML5/CSS grid frontend with live Tableau cloud rendering, it allows doctors to apply dynamic cohort filters instantly without page reloading or application sync failures.
4.	Social Impact / Customer Satisfaction	Healthcare professionals gain an accelerated, visually intuitive method for spotting compounding cardiovascular risks before critical cardiac events occur. This reduces clinical cognitive fatigue, shortens diagnostic validation timelines, and ultimately drives faster, data-backed preventative treatment plans—vastly improving overall patient care outcomes and hospital efficiency.
5.	Business Model (Revenue	The solution can be licensed as a B2B Software-as-a-Service (SaaS) platform for hospitals and diagnostic clinics on a tier-based subscription. Additional

Model) revenue can be generated through custom EHR (Electronic Health Record)

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		integration, premium bespoke dashboard engineering for specialized medical departments, and secure on-premise deployment services.
6.		The Flask web application architecture is highly modular. It can seamlessly ingest much larger clinical datasets, integrate live patient monitoring APIs, Scalability of and scale to track other chronic illnesses (e.g., respiratory or neurological the Solution conditions) simply by updating the cloud Tableau data engine, all without requiring significant changes to the existing backend web routing.

Prepared as part of the Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease capstone project.

